5.2 Power of Indices

5.11 Evaluate

a) $(1+2^3)^2$

b) $\frac{2^3-1}{2^2+3}$

Solution

a) $(1+2^3)^2$

- $1+2^3 = 1+8 = 9$
- $(1+2^3)^2 = 9^2 = 81$

b) $\frac{2^3-1}{2^2+3}$

- $\frac{2^3-1}{2^2+3}$
- $2^3 - 1 = 8^2 - 1 = 7$
- $2^2 + 3 = 2^2 + 3 = 7$
- $\frac{7}{7} = 1$

5.12 Simplify

$x^2y^2 + 3(xy)^2$

Solution

- The bracketed expression is simplified first

$(xy)^2 = (xy)(xy) = x^2y^2$

- So we have

$x^2y^2 + 3(xy)^2 = x^2y^2 + 3x^2y^2$

- $x^2y^2 + 3(xy)^2 = 4x^2y^2$

6.1 The laws of indices

- First law
  $a^m \times a^n = a^{m+n}$
- Second law
  $\frac{a^m}{a^n} = a^{m-n}$
- Key point: any number raised to the power 0 equals 1, that is $a^0 = 1$
- Third law
  $(a^m)^n = a^{mn}$
- Key point: $(a^m b^n)^k = a^{mk} b^{nk}$

6.2 Negative powers

- key poits - $a^{-m} = \frac{1}{a^m}$, $a^m = \frac{1}{a^{-m}}$

6.3 Square roots, cube roots and fractional powers

- Key point: $x^{\frac{1}{2}} = \sqrt{x}$, $x^{\frac{1}{3}} = \sqrt[3]{x}$
- Key point: $x^{\frac{1}{n}} = \sqrt[n]{x}$
- Key point: $a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$

6.4 Multiplication and division by powers of 10

6.5 Scientific notation

7. Simplifying algebraic expressions

7.1 Addition and substraction of like terms

7.2 Multiplying algebraic expressions and removing brackets

- Key point
  When multiplying
  positive x positive = positive
  positive x negative = negative
  negative x positive = negative
  negative x negative = positive

7.3 Removing brackets from a(b+c), a(b-c), (a+b)(c+d) and (a+b)(c-d)

- Key point
  $a(b+c) = ab + ac$
  $a(b-c) = ab - ac$
- Key point
  $(a+b)(c+d) = ac + bc + ad + bd$

8. Factorisation

8.1 Factors and common factors

8.2 Factorising quadratic expressions

- Key points

An expression of the form $ax^2 + bx + c$, where a, b and c are numbers, is called a quadratic expression. The coefficient of x^2 is a, the coefficient of x is b, and the constant term is c

- Quadratic expression where the coefficient of x^2 is 1

8.3 Different of two squares

$(x+m)(x+n) = (x+m)x + (x+m)n$
$x^2 + mx + nx + mn$
$x^2 + (m+n)x + mn$

- Expressions of the form $a^2 - b^2$ are known as the difference of two squares

9. Algebraic fractions

9.1 Introduction

When dividing

- $\frac{positive}{positive} = positive$
- $\frac{negative}{positive} = negative$
- $\frac{positive}{negative} = negative$

- $\frac{negative}{negative} = positive$

9.2 Cancelling common factors

When simplifying an algebraic fraction only factors common to both the numerator and denominator can be cancelled.

9.3 Multiplication and division of algebraic fractions

$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$

9.4 Addition and subtraction of algebraic fractions

9.5. Partial fraction

Let d = highest power of the denominator
Let n = highest power of the numerator
Then
if n ≥ d the fraction is improper
if n < d the fraction is proper